

Multi-AGN Monte Carlo Jet Power Batch

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PAPER_926: Multi-AGN Monte Carlo Jet Power Batch

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-11 **Session:** 211 **Source:** SCm phonon gap implementation (agn_{jet_power_curves}.py) **Calculator:** MultiAGNJetPowerMonteCarloBatchCalc (CP4 #510) **CVW:** v2.0.0 compliant

Abstract

Monte Carlo sampling (10^6 samples) of phonon-enhanced jet power P_{jet} across four AGN archetypes: 3C273-type ($M = 8.86e8 M_{\text{sun}}$), CenA-type ($5.5e7 M_{\text{sun}}$), TON618-type ($6.6e10 M_{\text{sun}}$), and TXS0506-type ($3e8 M_{\text{sun}}$). Γ draws from $N(\mu_{\Gamma}, \sigma_{\Gamma}^2)$ with $\mu = 1.25 \text{ THz}$, $\sigma = 0.15 \text{ THz}$ to model thermal phonon linewidth fluctuations. Reports mean, standard deviation, median, p5, and p95 percentiles of $P_{\text{jet}}(\Gamma)$ for each system. The P_{BZ} baseline scales as M^2 , giving a dynamic range of $\sim 10^8$ in base jet power across the four systems.

1. Core Equations

Section A: Lagrangian

$$P_{\text{jet}} = P_{\text{BZ}} * (1 + M_{\text{jet}}(\Gamma))$$

$$M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} * \exp[-(\Gamma - \Gamma_0)^2 / (2 * \sigma_{\Gamma}^2)]$$

$$P_{\text{BZ}} = (\pi / (6 * \mu_0)) * B^2 * r_g^2 * c * a^2$$

$$\Gamma \sim N(1.25 \text{ THz}, 0.15^2 \text{ THz}^2)$$

Section B: VDS/DVP/BH Number Systems

MC statistics: $\text{mean}(P_{\text{jet}})$, $\text{std}(P_{\text{jet}})$, $\text{median}(P_{\text{jet}})$, P_5 , P_{95}
Modulation range: $[\min(1+M_{\text{jet}}), \max(1+M_{\text{jet}})]$ over all draws
System diversity: $P_{\text{BZ}}(\text{TON618}) / P_{\text{BZ}}(\text{CenA}) \sim (6.6e10/5.5e7)^2 \sim 1.4e6$

Section SM: SM Anchors

3C273 – type : $M = 8.86e8 M_{\text{sun}}, \text{high-luminosity quasar}$
CenA – type : $M = 5.5e7 M_{\text{sun}}, \text{nearby radiogalaxy}$
TON618 – type : $M = 6.6e10 M_{\text{sun}}, \text{ultramassive quasar}$
TXS0506 – type : $M = 3e8 M_{\text{sun}}, \text{blazar neutrino source}$

2. Parameters

Parameter	Default	Description
$M_{\text{bh_Msun}}$	8.86e8	BH mass (solar masses)
a_{spin}	0.9	Spin parameter
B_{field}	50 T	Magnetic field
A_{jet}	1.5	Modulation amplitude
$\sigma_{\text{Gamma_THz}}$	0.08	σ_{Gamma}
$\Gamma_{\text{mean_THz}}$	1.25	MC mean Γ
$\Gamma_{\text{std_THz}}$	0.15	MC std dev
n_{samples}	100000	MC sample count

3. Key Results

System	P_{BZ} (erg/s)	(erg/s)	Spread
3C273-type	$\sim 10^{46}$	$\sim 2.5 * P_{\text{BZ}}$	+/-30%
CenA-type	$\sim 10^{42}$	$\sim 2.5 * P_{\text{BZ}}$	+/-30%
TON618-type	$\sim 10^{50}$	$\sim 2.5 * P_{\text{BZ}}$	+/-30%
TXS0506-type	$\sim 10^{44}$	$\sim 2.5 * P_{\text{BZ}}$	+/-30%

4. Physical Interpretation

The MC approach captures the stochastic nature of phonon linewidth fluctuations in real AGN environments. The 30% spread in P_{jet} at fixed BH mass and spin arises entirely from phonon frequency variations, providing a natural

explanation for AGN jet power variability on timescales of hours to days (matching observed VHE flaring). The consistent $\sim 2.5\times$ mean enhancement across all four systems demonstrates that phonon modulation is mass-independent, affecting only the multiplicative factor while P_BZ provides the mass-dependent baseline.

5. References

- PAPER_920: Monte Carlo jet power sampling (Session 210c)
 - PAPER_925: Quasar jet phonon modulation $M_{\text{jet}}(\text{Gamma})$
 - agn_{jet_power_curves}.py: 3-class standalone module
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Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_{Bi_i}\}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M-\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

$M-\sigma$ correction (PAPER_1048): The phonon-corrected $M-\sigma$ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[SSq]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	Fundamental

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Phonon frequency	$2\pi \times 1.25 \text{ THz}$ (Pd-D lattice)	Measured Pd-D phonon spectrum	Fukai (2005)	Mapped to SCm
ω_{SCm}				
Vacuum energy ρ_{vac}	$7.09 \times 10^{-37} \text{ kg/m}^3$	$\rho_{vac} \sim 10^{-29} \text{ g/cm}^3$	Planck 2018	Novel SCm scale
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: UQFF phonon-mediated vacuum coupling provides testable predictions beyond SM for this system.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: SCm-phonon (lattice resonance)

§A.2 Lagrangian Density

$$\mathcal{L}_{SCm_phonon} = \sum_{i=1}^{26} [U_{g,i} + U_{m,i} + U_{A,i} - U_{b,i}] \cdot S_{26}([SSq]) \cdot \Phi_{1.25\text{THz}}(\omega, \Gamma)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_m u \frac{\partial \mathcal{L}}{\partial (\partial_m u \phi)} = 0 \implies F_{U, Bi_i} = -\nabla U_{\text{eff}} + \Phi \cdot S_{26} \cdot E_{\text{net}}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms $\rightarrow$ SCm vacuum $\rightarrow$ phonon ω_{SCm} $\rightarrow$ lattice resonance $\rightarrow$
 F_{U, Bi_i} unified force $\rightarrow$ observational prediction

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{U_Bi_i}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{U_Bi_i}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1069	VDS-DVP-BSH Hybrid Calculator Unified

15 cross-reference(s) identified.

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